Copyrighted Materials
Copyright © 1994 Oxford University Press Retrieved from www.knovel.com

10

GENERAL ISSUES RELATED TO DIRECT SIMULATIONS

10.1 Introduction

The remaining chapters of this book are concerned with the detailed procedures of the DSMC method, the implementation of these procedures through demonstration programs, and the verification and illustration of the method through the results from these programs. The recommended DSMC procedures have evolved over a considerable period of time and this developmental process will continue. Alternative procedures are available for most aspects of the simulation, and there may be some argument over the best choice for particular applications. The issues that relate directly to the procedures will be dealt with as they arise in the chapters that follow. However, there are a number of general issues that are relevant to all direct simulation methods, and these are best discussed prior to the presentation and implementation of the DSMC algorithms.

10.2 The relationship to the Boltzmann equation

The Boltzmann equation dates from 1872 and, for the first ninety years from that date, theoretical advances in all regimes of fluid mechanics were based on the mathematical analysis of the relevant equations. Digital computers became generally available around 1960, and this introduced the capability for making direct physical simulations of gas flows without any recourse to the conventional mathematical models of the flow. However, the mathematical tradition in fluid mechanics was such that there were difficulties in gaining acceptance for the simulation methods. Tradition remains strong in some quarters and, in these, a numerical solution of the approximate mathematical model that is provided by the Boltzmann equation would be valued more highly than the results from a physical simulation that did not make the approximations that are inherent to the Boltzmann model. While progress has been made in methods for the direct numerical solution of the Boltzmann equation, it is clear that solutions are more readily obtained through direct simulation. The relative advantage of the simulation methods increases with the complexity of the flow and, for most cases of engineering interest, there is no practical alternative to direct simulation. Nevertheless, the simulation methods continue to suffer criticism for having a physical rather than a mathematical foundation.

In reply to this criticism, it can be pointed out that the Boltzmann equation was not 'handed down carved in stone'. It was derived through physical reasoning that is very close to that behind the DSMC procedures. As noted earlier, the assumptions of molecular chaos and a dilute gas are required by both the Boltzmann formulation and the DSMC method, and the major difference between them is that the DSMC method does not depend on the existence of inverse collisions. It has been shown (Bird 1970) that the Boltzmann equation can be derived through the DSMC procedures. While this derivation was based on the now obsolete 'time-counter' procedure for collision sampling, the new procedures (see §11.1) are even closer to classic kinetic theory. The DSMC method is, therefore, as soundly based as the Boltzmann equation itself. Moreover, because it does not rely on inverse collisions, it can be applied to complex phenomena such as ternary chemical reactions that are inaccessible to the Boltzmann formulation.

Just as there are computational approximations associated with direct numerical solutions of the Boltzmann equation, there are computational approximations inherent in the DSMC method. These are discussed in §10.5 and relate primarily to the number of simulated molecules and the finite intervals in space and time. It is these computational approximations, rather than the actual simulation procedures, that introduce the need for test cases to verify the method. While these test cases deal with particular flow configurations, the DSMC procedures are general. In addition, there are physical approximations associated with the molecular models and boundary interactions. With regard to these, it is much easier to introduce more complex and accurate models into the direct simulation environment than into the formal Boltzmann solutions.

The MD method is most appropriately used to study dense gas effects near the critical point. The Boltzmann equation is not valid for dense gases, and there is no equation that can be put in its place. To limit direct simulation objectives to a solution of the Boltzmann equation is unduly restrictive. The MD method was more readily accepted than the DSMC method, and it is probable that this was partly due to the absence of an accepted mathematical model at the particle level for dense gases and liquids. Another probable reason for this more ready acceptance is that the MD method is largely deterministic.

Given the physical basis of the DSMC method, the existence, uniqueness, and convergence issues that are important in the traditional mathematical analysis of equations, are largely irrelevant. At the same time, some people would obviously find it comforting if it was possible to construct an analytical representation of all of the procedures of the method, such that this representation did allow formal proofs of existence, uniqueness, and convergence. Unfortunately, the attempts to do this have generally proved to be inimical to the quality of the simulation. A problem that is common to all of these attempts is that many of the important practical details, such as the use of sub-cells, are extremely difficult to incorporate into an analytical model. Conditional steps, such as the action that is taken with

regard to collisions when there are just one or two molecules in a cell, pose even greater difficulties. In addition, some of the probabilistic aspects of the DSMC method, such as the undesirable introduction of random walk effects, appear to have posed insuperable difficulties and, while they may be present in the simulation, they are absent from the analytical representations. Finally, the procedures for physical effects such as chemical reactions and thermal radiation are sufficiently complex to be beyond any possible analytical representation. Any requirement for such a representation is an avoidable constraint that tends to inhibit the development of optimal simulation procedures.

Nanbu (1980) introduced a variation of the DSMC method that was derived directly from the Boltzmann equation, and a convergence proof for it has been presented by Babovsky and Illner (1989). Since the dependent variable in the Boltzmann equation is the velocity distribution function for a single molecule, the Nanbu procedures alter the velocity for only one of the two molecules that are involved in each collision. This is clearly physically inconsistent with a real gas, and has the obvious disadvantage that twice as many collisions must be calculated. More seriously, there is no longer exact momentum and energy conservation in each collision and, while they are conserved on the average, there will be random walks in these quantities at each collision. Random walks, which are discussed more fully in §10.4, degrade the results for large perturbation problems and can render useless the results for flows with small perturbations. The 'Finite Pointset Method' (Gropengiesser et al. 1991) is another variant of the DSMC method that is claimed to be directly related to the Boltzmann equation. Both the location and velocity of the simulated molecules appear to be rounded to a finite number of points at predetermined locations in phase space. The scatter in the results seems higher than would normally be expected for DSMC calculations of similar magnitude, and the most probable explanation for this is that the rounding procedure also leads to random walks.

10.3 Normalized or dimensional variables

Basic physical data is always presented in a dimensional form, but the results from experiments are generally expressed in a dimensionless form. The latter is particularly desirable when the dimensionless parameters are those that can be used to correlate the data from many different experiments. In analytical work, the basic equations are often transformed into a dimensionless form so that the results are automatically in the most general form. The early applications of the DSMC method were to basic flows and employed the unrealistic hard sphere molecular model. This meant that the results could not be compared directly with experimental values, but could be readily expressed in terms of the dimensionless parameters. The desired comparison with experiment could then be made by matching the dimensionless parameters. The Knudsen number is the most important parameter for rarefied gas flows, and this was specified as

one of the data items. There was then a strong case for expressing the rest of the data in a dimensionless or normalized form.

The programs in the earlier exposition (Bird 1976a) of the DSMC method employed normalised variables, but a number of problems became apparent over the years. One was that some readers had more difficulty in understanding the normalization procedures than they had with the other DSMC procedures. More seriously, some of the normalization procedures gave a misleading impression of the basis of the method. In particular, the specified mean free path was obtained through the setting of a 'cross-section' variable that was proportional to the number of simulated molecules. Like the diameter in the MD method, the diameter based on this variable was almost always many orders of magnitude larger than the real molecular diameter. This apparent size was of no significance in the DSMC method, because it was used only to allow for the comparatively small number of the simulated molecules when setting the collision rate. It is preferable to specify the number of molecules that are represented by each simulated molecule as a distinct parameter, and to employ the real gas cross-section when setting the collision rate. A further problem is that it is not possible to obtain a closed form expression for the mean free path in a mixture of VHS or VSS molecules and normalization is then extremely difficult . Also, complex flows may include chemical reactions and normalization then becomes almost impossible.

An additional factor is that most DSMC calculations are now made in an engineering context or in parallel with experiments. The molecular models are now sufficiently realistic for the simulation results to be compared directly with the measured values. It is therefore desirable for all the variables in the programs to be in a dimensional form. If dimensionless results are required, the normalization is best applied to the results. This simplifies many of the procedures in that the physical constants, such as the Boltzmann constant, appear explicitly in the programs. The demonstration programs in the following chapters employ dimensional variables, with SI units used throughout.

10.4 Statistical scatter and random walks

The representation of the extremely large number of real molecules by a relatively small number of simulated molecules means that the DSMC results are subject to a level of statistical scatter that is, in most applications, many orders of magnitude greater than the scatter that is present in the real gas. This scatter can generally be assumed to follow the Poisson distribution, and the standard deviation is of the order of the inverse square root of the sample size. The total number of simulated molecules is generally such that the level of scatter associated with an instantaneous sample is unacceptably large, and a cumulative sample is required. The results are generally based on a time average for a steady flow or an ensemble average for an unsteady flow.

An important point is that the volume of physical space is completely specified only when the flow is three-dimensional. The width of the 'slice' of flow in a two-dimensional flow, or the 'cross-sectional area' of a one-dimensional flow may be assigned in an arbitrary manner. They are usually set to unit value for convenience, and this generally leads to an extremely large value for the number of real molecules that are represented by each simulated molecule. However, the width or cross-section can be set to the extremely small values that correspond to a one-to-one correspondence between the real and simulated molecules. The scatter should then have real physical significance, and this situation has been studied and reported in a number of papers that have been reviewed by Garcia (1990). The initial results (Garcia 1986) showed that the DSMC method exactly reproduced the equilibrium fluctuations and, in a non-equilibrium system, the peak of the measured density–velocity static correlation function was linearly proportional to the temperature gradient, in agreement with theory and light scattering experiments. Later studies (Malek Mansour et al. 1987) showed that the static correlation of the temperature fluctuations was in excellent agreement with the numerical solution of the correlation equations. Comparison was also made (Garcia et al. 1987) of the fluctuations in a Couette flow simulation with the predictions of fluctuating hydrodynamic theory, and the agreement was again very good.

Advantage can be taken of the fact that the simulated molecules almost always represent a much larger number of real molecules to impose uniform rather than random spacing in physical and velocity space between variables in the situations in which it is physically appropriate. DSMC implementations have generally taken advantage of this with regard to the initial locations of the molecules. A so-called 'low discrepancy' variation of the DSMC method was introduced (Babovsky et al. 1989) in which known uniformities were applied to the distributions in all procedures. For example, in the case of VHS molecules, the post-collision direction of the relative velocity in the centre-of-mass frame of reference is uniformly distributed in space. Therefore, for all the collisions in a cell during a particular time step, these directions could be spread out as evenly as possible in space. One option for doing this is through the replacement of completely random sequences by quasi-random or low discrepancy sequences (Kuipers and Niederreiter 1976). Morokoff and Caflisch (1991) have shown that the gains from these procedures are very small unless the instantaneous sample population of the distribution is sufficiently large to provide a reasonable representation of that distribution. Unfortunately, most of the distributions that involve molecular velocities are multi-dimensional in velocity space and deal with a small sample population at each time step. Systematic variance reduction has not been demonstrated for the DSMC method. However, it was shown in §7.7 that some degree of variance reduction can be obtained when this principle is applied to the much larger samples that can be employed in the test particle Monte Carlo method for collisionless flows.

As noted above, the statistical fluctuations decline with the square root of the sample size and it should be possible to achieve any desired level of accuracy by continuing or repeating the simulation to build up the size of the sample to the required magnitude. The gradual approach of a sampled quantity to its correct value is sometimes referred to as 'convergence'. This is undesirable because the progressive reduction in statistical scatter is a completely different process to that commonly referred to as 'convergence' in the conventional CFD methods.

Some implementations have produced results in which the scatter fails to decline beyond a certain level, or may even increase with time. This behaviour has been found to be due to simulation procedures that allow a random walk in one or more of the variables. One characteristic of a random walk is that the displacement from the mean position or value *increases* with the square root of time. This displacement may cause an output variable to either increase or decrease, and the mean time between zero crossings also increases with the square root of time.

Random walks can arise whenever one of the molecular quantities are conserved only on the average, rather than exactly, in any of the simulation procedures. The molecular quantities are the position coordinates, the velocity components, and the internal energies. The random walk due to the 'Nanbu procedure' has already been discussed in §10.2. The application of species based *weighting factors* leads to similar random walks. Such weighting factors have been used when dealing with trace species, because these would be represented by a very small sample if the number of real molecules represented by each simulated molecule was the same for each species. The sample of the trace species is increased if each of the simulated molecules of this species represents fewer real molecules than those of the dominant species, and this difference is termed a weighting factor. However, in a collision between molecules with different weights, the properties of the molecules of the species with the higher weight must be changed in only some of the collisions. Momentum and energy are then not conserved exactly in each collision, but only in the average, and each departure from exact conservation is effectively a random walk in that quantity. The deviation due to the random walk can more than outweigh the reduced scatter in the trace gas properties due to the increased sample, and the use of species based weighting factors is no longer recommended.

Random walks also arise when the position coordinates or the velocity components are rounded-off in order to be stored as discrete values rather than as exact numbers. The motivation for this can be to reduce the memory storage requirements by 'packing' the coordinates and velocity components into integers. The discrete ordinate methods generally avoid this problem with regard to the velocity components by restricting the collisions to a discrete and pre-determined number that exactly conserve momentum and energy. This ceases to be possible when the internal energy modes are included. There does not appear to be any way of avoiding random walks when position coordinates are rounded to discrete values, even when there are an extremely large number of available sites.

Because of the limited precision with which variables are manipulated and stored in computers, some degree of rounding is unavoidable. This means that are always random walks, albeit with a very small step length, even when the logic is 'exact'. This effect was investigated by monitoring the total energy in a DSMC simulation of a uniform gas with a fixed sample of 1,000 molecules. After 50,000,000 collisions (or 100,000 per molecule) with a 32 bit word length, the total energy was found to change by no more than three parts in a thousand. This indicates that the unavoidable random walks are of acceptable magnitude.

The unavoidable statistical scatter, and the fact that it declines only as the square root of the sample size, has been a major problem for the DSMC method. The magnitude of the computational task meant that, for many years, the method was thought to be applicable only to problems that involve large perturbations. For example, the scatter applies to the molecular velocity components so that, in a problem that involves low subsonic stream velocities, the statistical 'noise' could be at least an order of magnitude greater than the required 'signal'. The reduction in the cost of computing has meant that it is now possible, at least for one-dimensional flows, to deal with time or ensemble averages in the millions, and the scatter is then reduced to less than one part in a thousand. A weak disturbance can be completely lost in the scatter in an instantaneous sample or during the earlier stages of a simulation but, as long as the procedures are such that random walks are precluded, it emerges correctly as the average builds up.

10.5 Computational approximations

The computational approximations associated with the DSMC method are the ratio of the number of simulated molecules to the number of real molecules, the time step over which the molecular motion and collisions are uncoupled, and the finite cell and sub-cell sizes in physical space.

The first approximation leads to the statistical scatter, as discussed in the preceding section, but there does not seem to be any value of this ratio at which the fluctuations change their nature or become unstable. It is possible to argue qualitatively that fluctuations may become unstable in some flow configurations. For example, consider a region within the bow wave in a low-density hypersonic shock layer. A positive fluctuation in density would lead to more collisions with the oncoming freestream molecules in this region, and the positive fluctuation would be expected to be reinforced. However, there have been many simulations of this class of flow and these have shown that the fluctuations are stable. This is an empirical result and it would be desirable to have an analytical proof of this stability. The fact that the scatter, when it does have physical significance, is consistent with the analytical predictions provides some reassurance.

The most serious statistical problem is when a significant effect in the real gas is a consequence of the few molecules towards the extremes of the

distribution. The number of simulated molecules may be so small in comparison with the real molecules that the important part of the distribution may not be adequately populated. This problem generally arises in connection with chemical reactions or thermal radiation. A solution in the latter case is to take advantage of the large number of real molecules represented by each simulated molecule and assign a distribution of electronic states, rather than a single state, to each simulated molecule. Of all the flows that have been simulated, the only non-reacting flow that has proved (Bird 1976b) to be sensitive to the number of simulated molecules is the hypersonic flow within a conical molecular beam skimmer. This is because a single molecule that has a sufficiently high velocity normal to the axis to strike the inside of the cone can produce such a large number of similar molecules when reflected back into the beam, that the hypersonic beam breaks down to a subsonic flow. The time to breakdown is much longer in the simulated flow than in the real flow. Note that this is essentially a physical effect, albeit a spurious one, that is predictable from the statistics.

The DSMC simulation becomes more exact as the time step and cell size tend to zero. It should be noted that disturbances can propagate at the sound or shock wave speed even though the ratio of the cell size to the time step may be much smaller than this ratio. There is, therefore, no stability criterion that is analogous to the Courant condition for finite difference methods in continuum CFD. During the early stages of the development of the method, some concern was expressed (Bird 1976a) about large values of this ratio. This was because the relative locations of the molecules in the cell were disregarded when choosing potential collision pairs. It was feared that coincidental collisions between molecules at opposite sides of the cells, coupled with the immediate migration of these molecules to the next cell, might cause false disturbances to propagate with speed equal to the ratio of the cell size to the time step. In, practice this ratio has been set to many times the speed of the sound or shock waves and there has been no sign of false disturbances. This is presumably because the predicted disturbances relied on continuance of unlikely coincidences. In any case, the recent adoption of sub-cells, as described below, has significantly reduced the scope for any such effects.

The DSMC method originally regarded the molecules in a cell as a representative sample of those in the real gas at the location of the cell. Then, as noted above, the relative positions of the molecules within the cell were disregarded when choosing potential collision pairs. Meiburg (1986) pointed out that if there is a velocity gradient across a cell, the choice of a collision pair from opposite sides of the cell can, when coupled with an adverse set of randomly selected impact parameters, actually reverse the portion of the systematic gradient that is represented by the pre-collision velocities of that pair. This particularly adverse case is unlikely, and it would have been more accurate to say that the collisions tend to reduce the vorticity within the cell. One could reply that most CFD methods do not allow vorticity *within* a cell and the vorticity can only be represented by

the velocity gradients *between* cells. To the extent that the DSMC method retains any vorticity in a cell, it is an improvement on most CFD methods. Fortunately, it is possible to do very much better through the introduction of sub-cells that cause all collisions to be between near neighbours. The DSMC procedures include an optimally efficient indexing scheme to allow pairs within a particular cell to be selected from an unsorted array of molecules. This has been extended to allow the choice to be from sub-cells that are formed by very small subdivisions of the cells. The use of sub-cells involves a negligible computational overhead, and the DSMC method has now been shown to provide an excellent representation of vortical flows. In fact, the original version is capable of providing good results for such flows, and Meiburg's contrary conclusions were based on a calculation with a cell size that was an order of magnitude larger than the recommended size.

The linear dimensions of the cells should be small in comparison with the scale length of the macroscopic flow gradients in the direction in which the dimension is measured. In regions with large macroscopic gradients, this generally means that the cell dimension should be of the order of one third the local mean free path. The time step should be much less than the local mean collision time. These guidelines will be subject to critical review in some of the examples in the following chapters.

In addition, both obvious and subtle errors can arise from the inescapable imperfections of computer arithmetic. These 'numerical artefacts' are discussed in Appendix E.

References

BABOVSKY, H., GROPENGIESSER, F., NEUNZERT, H., STRUCKMEIER, J. and WIESEN, B. (1989). Low-discrepancy method for the Boltzmann equation, *Prog. in Astro. and Aero.* **118**, 85–99.

BABOVSKY, H. and ILLNER, R. (1989). A convergence proof for Nanbu's simulation method for the full Boltzmann equation, *SIAM J. Num. Anal.* **26**, 45–65.

BIRD, G.A. (1976a). *Molecular gas dynamics*, Oxford University Press.

BIRD, G.A. (1976b). Transition regime behaviour of supersonic beam skimmers, *Phys. Fluids* **19**, 1486–1491.

BIRD, G.A. (1970). Direct Simulation of the Boltzmann Equation, *Phys. Fluids* ,**13** 2676–2681.

GARCIA, A.L. (1990). Hydrodynamic fluctuations and the direct simulation Monte Carlo method. In *Microscopic simulations of complex flows*, (ed. M. Mareschal), pp 177–188, Plenum Press, New York.

GARCIA, A. (1986). Nonequilibrium fluctuations studied by a rarefie-gas simulation. *Phys. Rev. A* **34**, 1454–1457.

GARCIA, A.L., MALEK MANSOUR, M., LIE, G.C., MARESCHAL, M., and CLEMENTI, E. (1987). Hydrodynamic fluctuations in a dilute gas under shear, *Phys. Rev. A* **36**, 4348–4355.

GROPENGIESSER, F., NEUNZERT, H., STRUCKMEIER, J., and WIESEN, B. (1991). Rarefied gas flow around a disc with different angles of attack, In *Rarefied gas dynamics*, (ed. A.E. Beylich), pp 546–553, VCH, Weinheim.

KUIPERS, L. and NIEDERREITER, H. (1976). *Uniform distribution of sequences*, Wiley, New York.

MALEK MANSOUR, M., GARCIA, A.L., LIE, G.C., and CLEMENTI, E. (1987). Fluctuating hydrodynamics in a dilute gas, *Phys. Rev. Letters* **58**, 874–877.

MEIBURG, E. (1986). Comparison of the Molecular Dynamics method and the direct simulation technique for flows around simple geometries, *Phys. Fluids* **29**, 3107–3113.

MOROKOFF, W.J. and CAFLISCH, R.E. (1991) A quasi-Monte Carlo approach to particle simulation of the heat equation, UCLA Computational and Applied Mathematics Report 91-13.

NANBU, K. (1980). Direct simulation scheme derived from the Boltzmann equation. I. Multicomponent gases, *J. Phys. Soc. of Japan* **45**, 2042–2049.